UNIVERSITÄT LEIPZIG

Morphology

Natural Language Processing
Leonie Weissweiler

RECAP

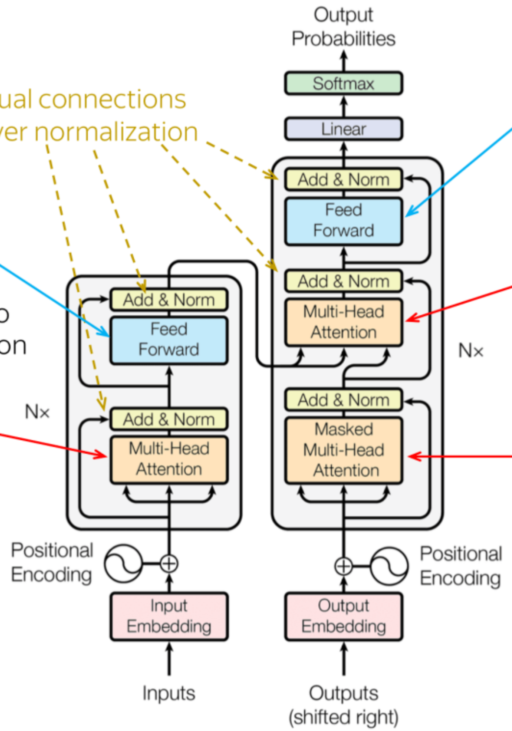
PUTTING IT ALL TOGETHER

Residual connections and layer normalization

Feed-forward network: after taking information from other tokens, take a moment to think and process this information

Encoder self-attention: tokens look at each other

queries, keys, values are computed from encoder states



Feed-forward network: after taking information from other tokens, take a moment to think and process this information

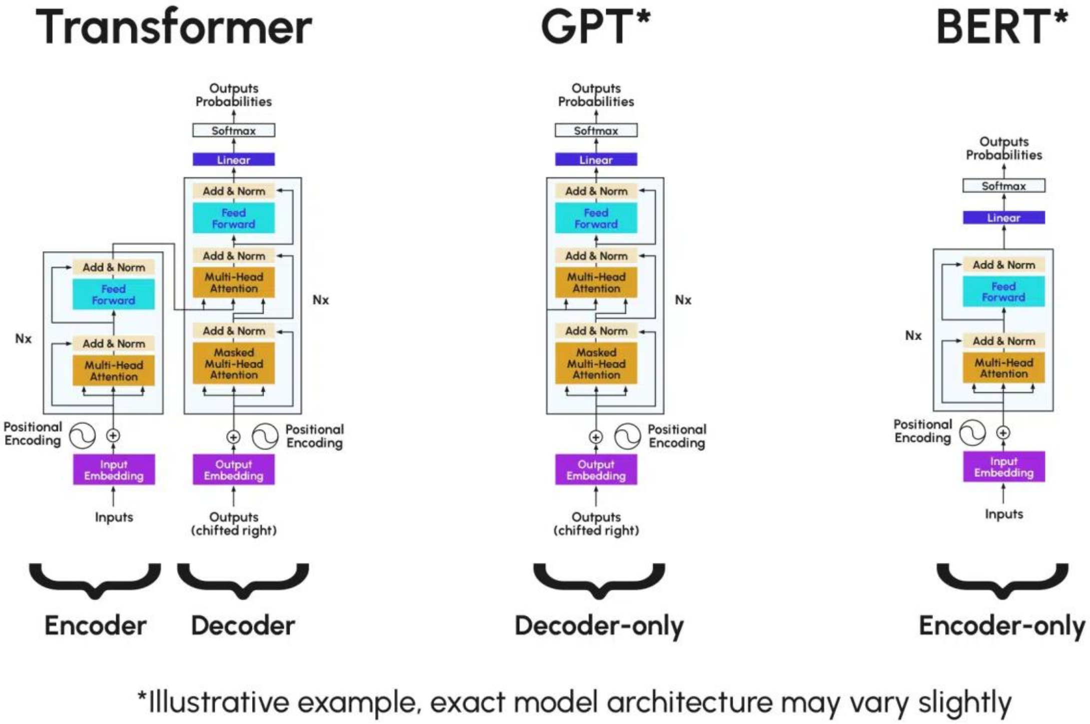
Decoder-encoder attention: target token looks at the source

queries – from decoder states; keys and values from encoder states

Decoder self-attention (masked): tokens look at the previous tokens

queries, keys, values are computed from decoder states

PREVIEW: ENCODER-ONLY AND DECODER-ONLY



IN-CONTEXT LEARNING

Any information about the task is directly in the prompt so that the model can learn from context, but the model is never updated!

BEAM SEARCH: CORE IDEA 💡

- On each step of decoder, keep track of the **k** most probable partial translations (called **hypotheses**)
  - k is the **beam size** (in practice around 5 to 10)
- A hypothesis has a **score:**

$$\text{score}(y_1, \ldots, y_t) = \log P_{\text{LM}}(y_1, \ldots, y_t | x) = \sum_{i=1}^{t} \log P_{\text{LM}}(y_i | y_1, \ldots, y_{i-1}, x)$$

  - Scores are all negative, and **higher score is better**
  - We search for high-scoring hypotheses, tracking top k on each step

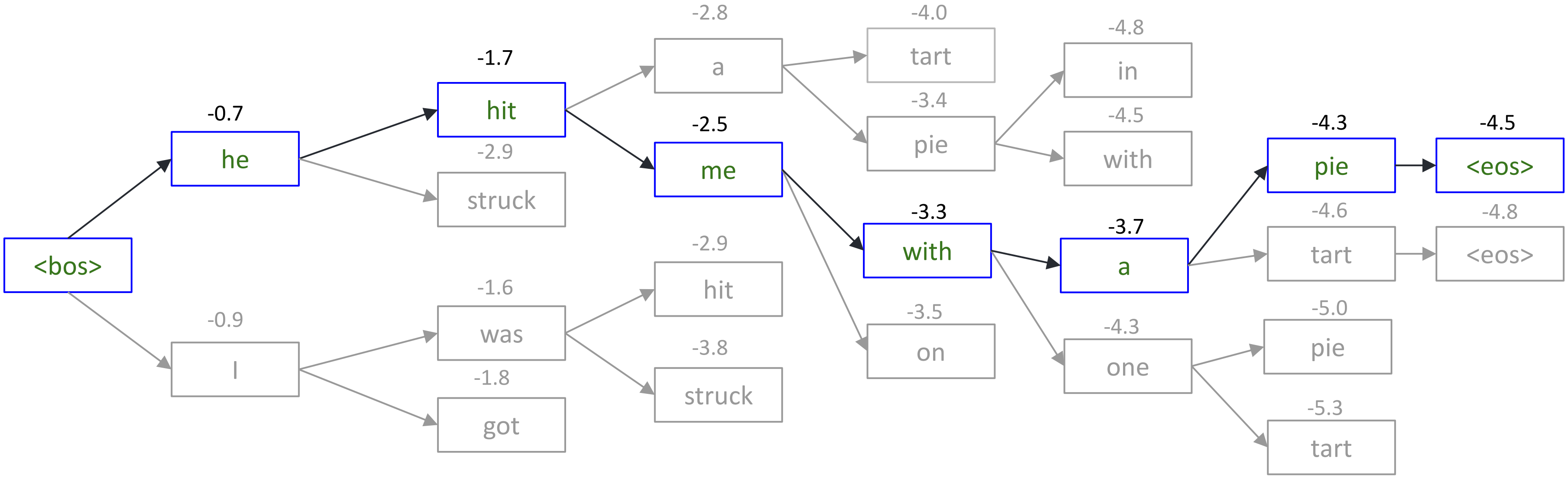
- Beam search is **not guaranteed** to find optimal solution
  - But it's efficient!

From: http://web.stanford.edu/class/cs224n/slides/cs224n-2021-lecture07-nmt.pdf

BEAM SEARCH: EXAMPLE

Beam Size = k = 2

$$\text{score}(y_1, \ldots, y_t) = \sum_{i=1}^{t} \log P_{\text{LM}}(y_i | y_1, \ldots, y_{i-1}, x)$$

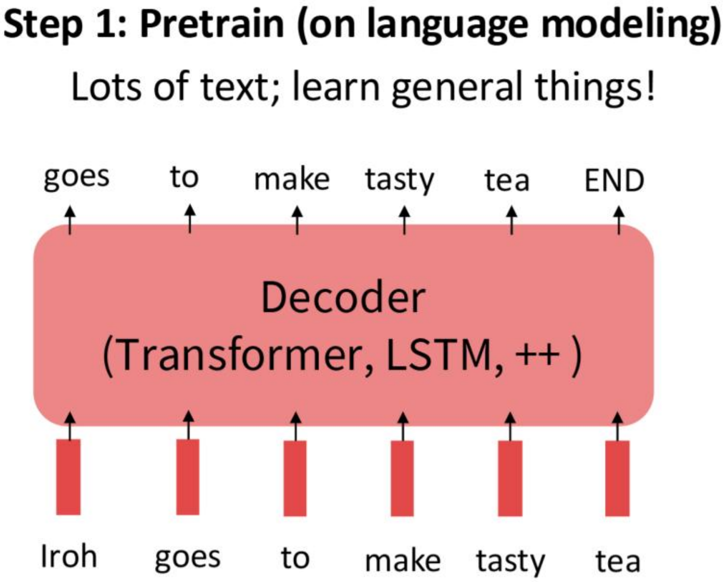


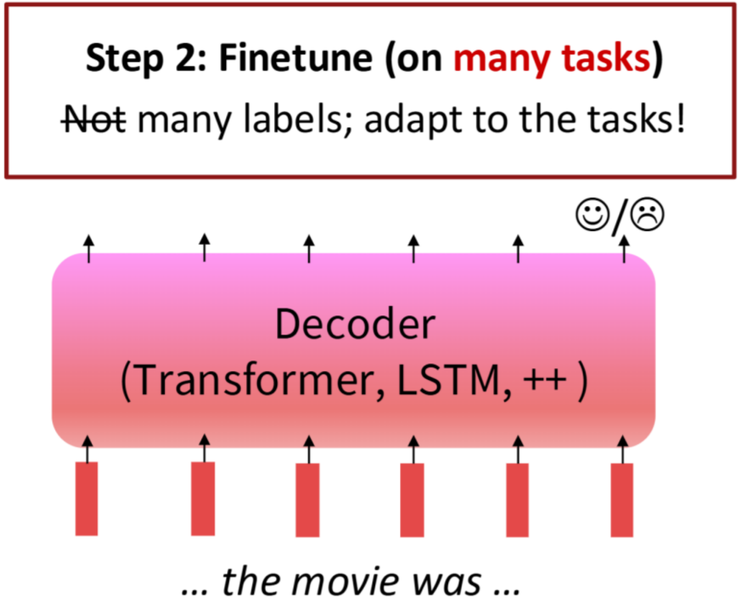
We found the top-scoring hypothesis!

BEAM SEARCH: STOPPING CRITERIA

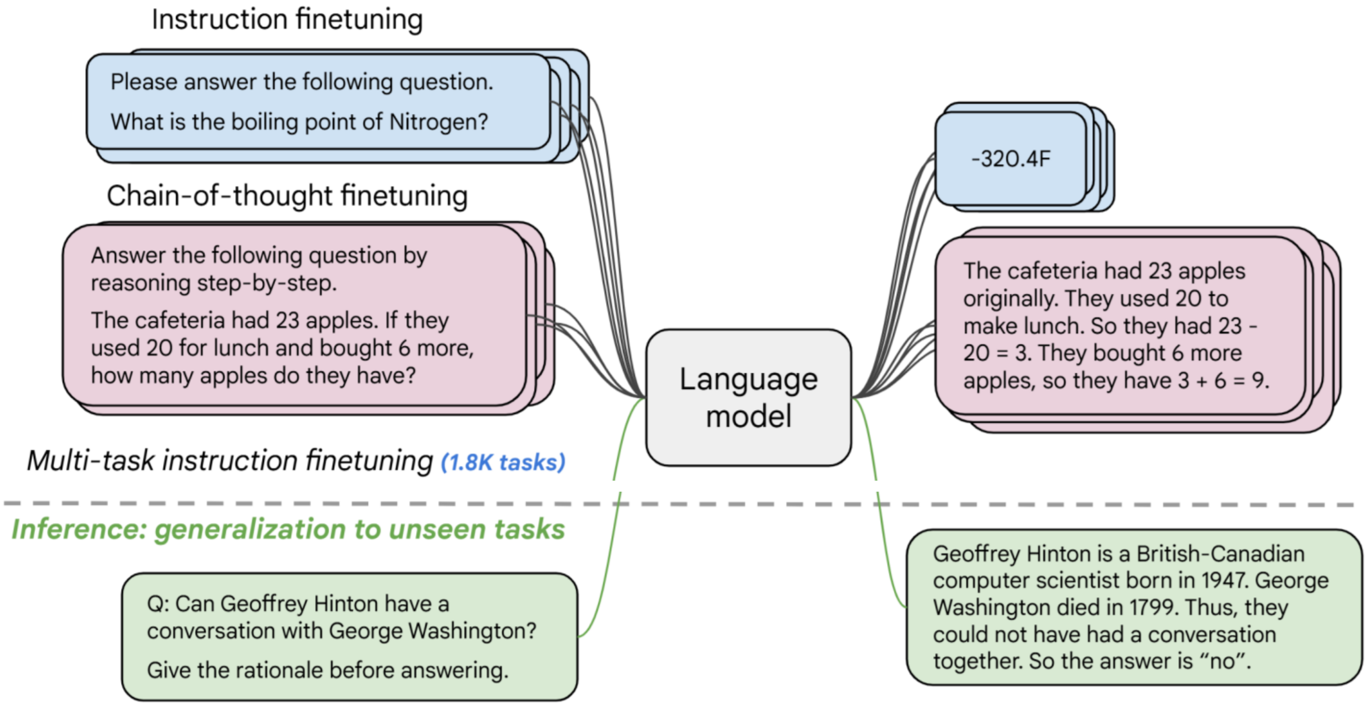
- Different hypotheses may produce `<eos>` tokens on different timesteps
  - When a hypothesis produces `<eos>`, that hypothesis is complete
  - Place it aside and continue exploring other hypotheses via beam search
- Usually we continue beam search until:
  - We reach timestep `T` (where `T` is some predefined cutoff), or
  - We have at least `n` completed hypotheses (where `n` is pre-defined cutoff)

FINETUNING FOR MANY TASKS AT ONCE



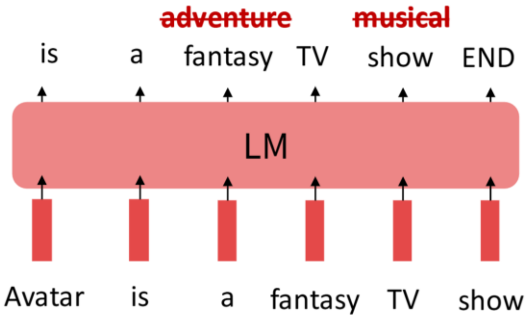


GENERALISATION TO UNSEEN TASKS



PROBLEMS WITH INSTRUCTION FINETUNING

- Problem 1: tasks like open-ended creative have no right answer.
  - E.g. Write me a story about a dog and her pet grasshopper.
- Problem 2: language modelling penalises all token-level mistakes equally, but some errors are worse than others



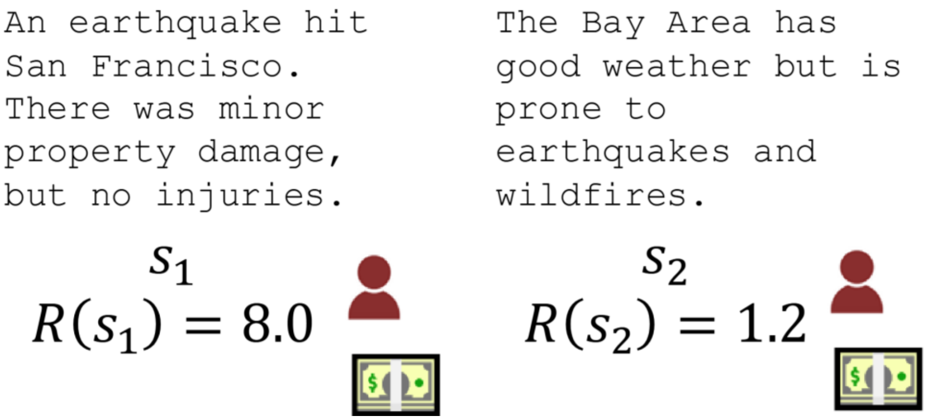
→ Mismatch between the LM objective and the objective of satisfying human preferences, even with instruction tuning

Big question: how can we satisfy human preferences explicitly?

HUMAN PREFERENCES AS REWARD

Problem 1: human-in-the-loop constantly providing reward signals is expensive!

Solution: instead of directly asking humans for preferences, model their preferences as a separate NLP problem!



→ Train a language model as Reward Model RM(s) to predict human Preferences from an annotated dataset, Then optimise for RM(s) instead of R(s)

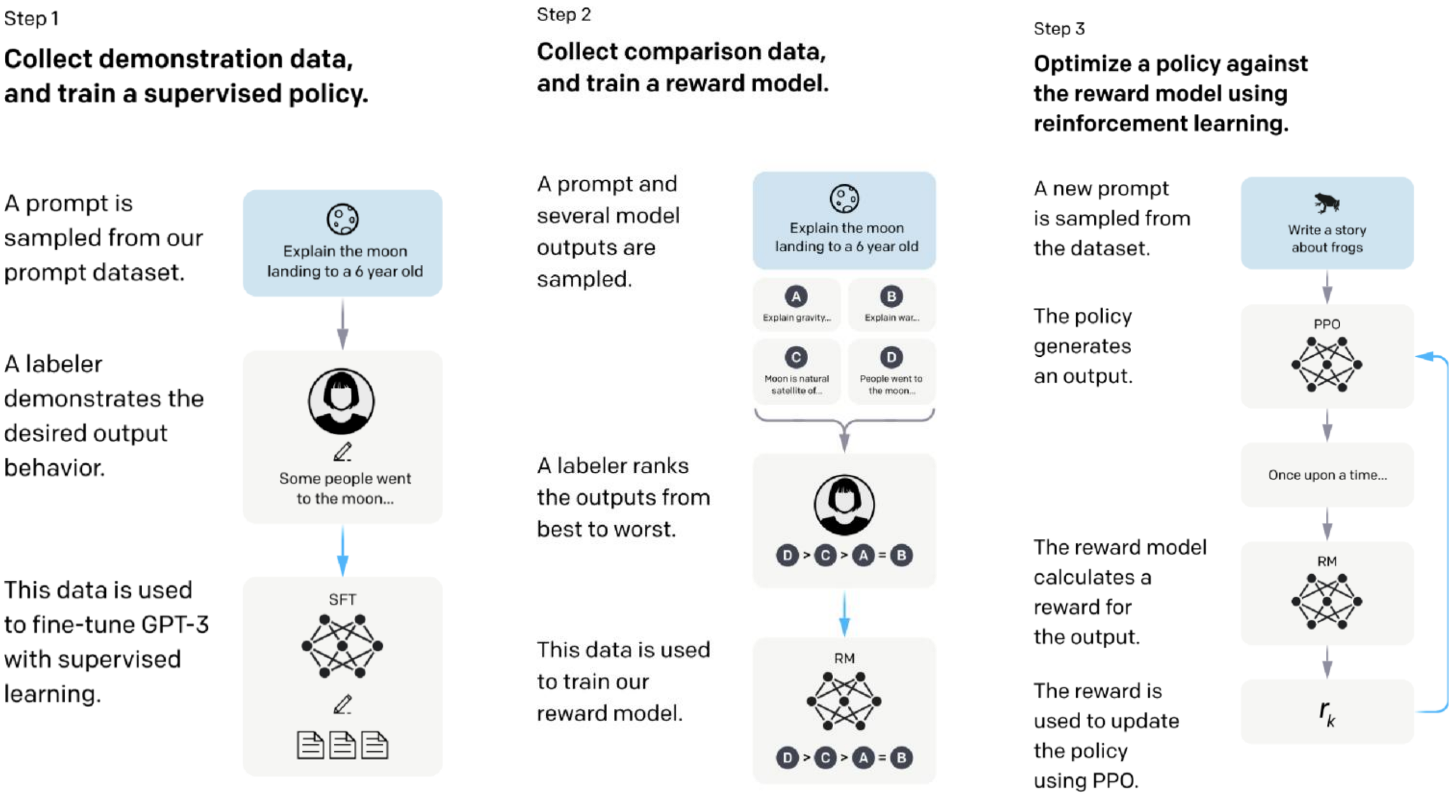
HUMAN PREFERENCES AS REWARD

Problem 2: human judgements are noisy and miscalibrated!

Solution: instead of asking for direct ratings, ask for pairwise comparisons, which are more reliable

s_1		s_3		s_2
An earthquake hit San Francisco. There was minor property damage, but no injuries.	>	A 4.2 magnitude earthquake hit San Francisco, resulting in massive damage.	>	The Bay Area has good weather but is prone to earthquakes and wildfires.

RLHF VISUALLY



CHATGPT

InstructGPT but

- transforming the dataset into a dialogue format
- Humans providing conversations where they played both sides, using model-written suggestions to compose their responses
- Extra preference data, created by selecting model-written messages, sampling alternative completions and having AI trainers rank them
- Fine-tuned using Proximal Policy Optimisation (a kind of RLHF, don’t worry)

PROBLEMS WITH HUMAN REWARDS

Human preferences are unreliable!

- Reward hacking is a common problem in RL
- Chatbots are rewarded to produce responses that seem authoritative and helpful, regardless of truth
- This can lead to making up facts and hallucinations


TECHNOLOGY

Google shares drop $100 billion after its new AI chatbot makes a mistake

February 9, 2023 · 10:15 AM ET

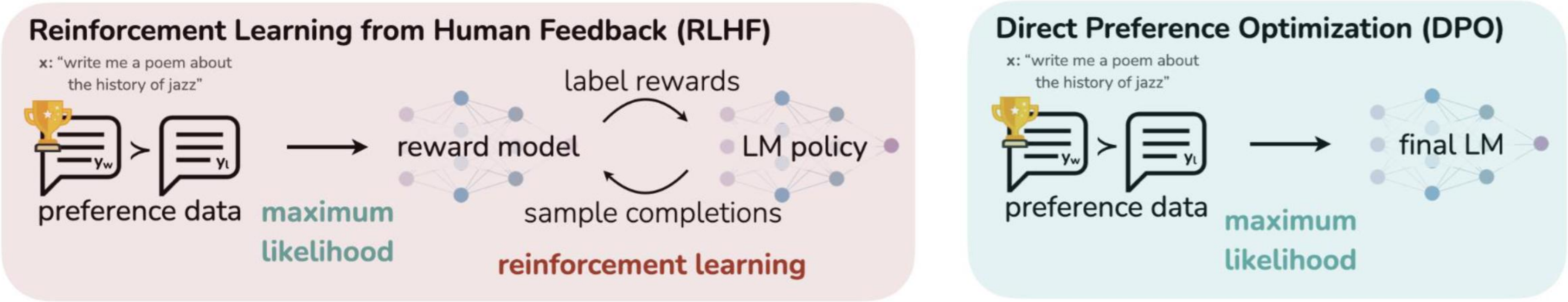
https://www.npr.org/2023/02/09/1155650909/google-chatbot--error-bard-shares

Bing AI hallucinates the Super Bowl

DIRECT PREFERENCE OPTIMISATION (DPO)

It turns out we didn't need the RL in the first place!

We still need preference data, but no more reward model



QUESTIONS?

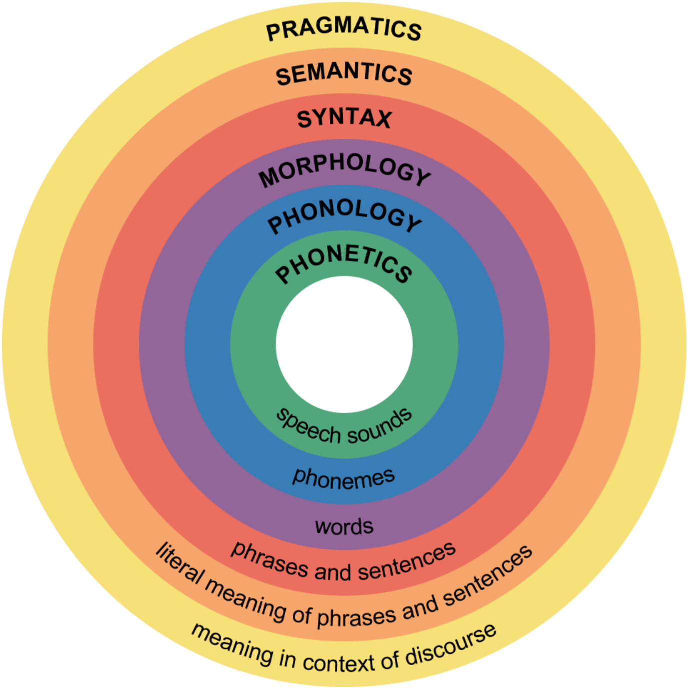
QUESTIONS?

LINGUISTICS

The scientific study of language

- Theoretical linguistics: understanding the fundamental nature of language
- Applied linguistics: utilising the findings about the nature of language for applications, e.g. better language teaching, or dictionaries
- Computational linguistics: computational modelling of natural language
- Empirical linguistics: using corpus data to understand language
- Historical linguistics: the study of how language changes over time
- Linguistic typology: classifies languages according to structural features

SUBFIELDS OF LINGUISTICS



- Levels of analysis of language and the respective subfields of linguistics
- Phonetics and phonology will not be covered here, because they require a different kind of LM not covered here

PLAN FOR THE REST OF THE SEMESTER

- Four lectures on linguistics: morphology, syntax, semantics, pragmatics. For each:
  - Purely linguistic introduction
  - (historical) NLP models, technologies, and applications
  - Intersection with LLMs
- One lecture on current topics in NLP: multilinguality, ethics, environmental aspects

OUTLINE FOR TODAY

- Morphology
- Computational Morphology
- Morphology and LLMs

QUESTIONS?

MORPHOLOGY

WORDS

- Words in natural languages encode many pieces of information
- What is the meaning of a word?
- How do words in a sentence interact with each other?
  - Subject-verb agreement
  - Adjective-noun agreement
  - …
- What lexical category does a word belong to?
  - Noun
  - Verb
  - …
- What can we say about the internal structure of a word?
  - Determine the parts a complex word is composed of
  - Specify morphological features such as number, gender, tense, …

INTERNAL STRUCTURE OF WORDS: EXAMPLE

- English: *I am swim-m-ing*
  - We know the meaning of *(to)* swim
  - -ing: marks the progressive form
  - Why the extra m?
- Turkish: *Ben yüz-üyor-um*

  I.Nom *swim*-Prog-1P.Sg
  - yüz means ‘swim’
  - -üyor corresponds to English *-ing*
  - -um indicates the person

  → Inflected Turkish verb contains more information

MORPHOLOGY

- The study of the internal structure of words
- Oldest sub-discipline of linguistics, e.g. well-structured lists of Sumerian words going back as far as 1600 BC
- The term morphology was invented in the 2nd half of the 19th century, by which point phonology and syntax had long been invented, so in that sense it is also a young discipline

MORPHEMES

- Morphemes are the smallest units in language that carry meaning
- E.g. “swimming” has two morphemes, “swim” and “-ing”
- Morphology studies
  - How morphemes combine
  - How morphemes function

ROOTS AND AFFIXES

- Some morphemes give words their basic meaning. These are called roots.
- Other morphemes are added to words to make new words, or to make new forms of existing words. These are called affixes.
- un-think-able; kitten-s
- Roots provide basic meaning and affixes modify it
- The roots (first column) express the basic meaning
- Affixes add grammatical meaning (second column) or modify the semantic meaning (third column)

<root>	<root>ing	<root>er
run	running	runner
think	thinking	thinker
program	programming	programmer
kill	killing	killer

BOUND MORPHEMES

Some roots are bound, which means they don’t occur as independent words.

Consider –ceive in

- Conceive
- Deceive
- Receive
- Perceive

The reason is sometimes unclear (etymology)

FOUR TYPES OF AFFIXES ATTACH TO BASES

- Bases are strings to which affixes can be applied
- Prefixes are added to the beginning of a base
- Suffixes are added to the end of the base
- Circumfixes are added to the beginning and end of the base simultaneously
- Infixes are inserted inside the base

PREFIXES AND SUFFIXES

Prefixes	regular		→	**ir**-regular	
	nuptial	‘wedding’	→	**pre**-nuptial	‘before wedding’
Suffixes	real	‘actual’	→	real-**ize**	‘become actual’
	hunt		→	hunt-**er**	

CIRCUMFIXES AND INFIXES

Circumfix	sammel*	‘gather’	→	**ge**-sammel-**t**	‘has gathered’
	light	‘light’	→	**en**-light-**en**	‘make light’
	bold	‘bold’	→	**em**-bold-**en**	‘make bold’
Infix	California		→	Cali-**freakin’**-fornia	
	sulat†	‘write!’	→	s**um**ulat	‘will write’

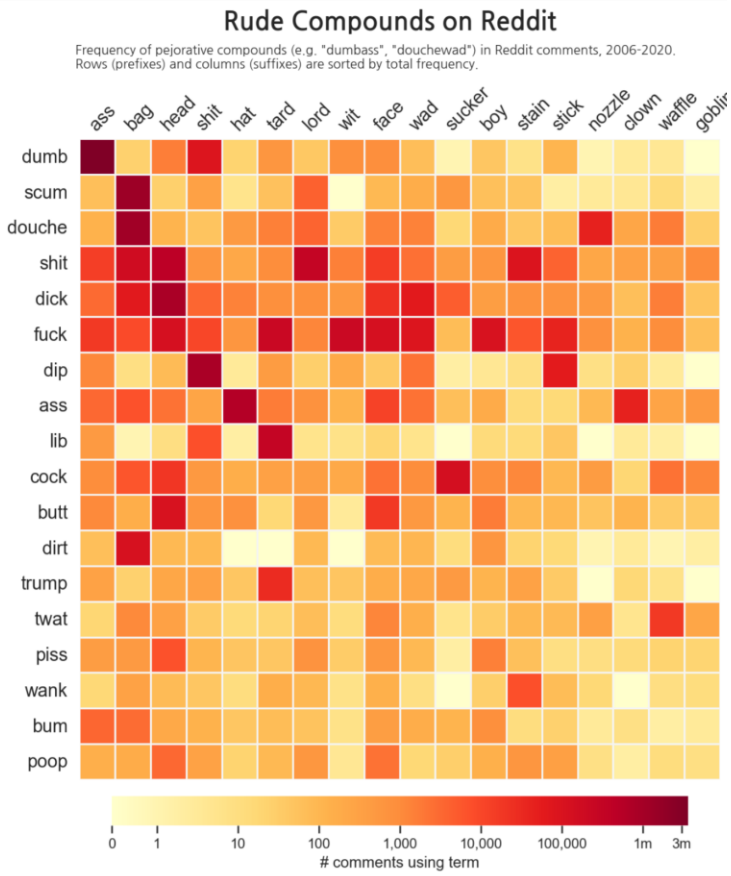
(German and Tagalog)

NON-CONCATENATIVE MORPHOLOGICAL OPERATIONS

There are morphological operations on bases other than affixation:

- Reduplication
  - sulat → susulat (Tagalog)
  - anak ‘child’ → anakanak ‘children’ (Indonesian)
- Apophony (umlaut, ablaut, etc.)
  - Foot → feet
  - Sing → sang, sung
- Transfixation (templatic morphology)
  - tb → kitab‡ ‘book’, kataba ‘he wrote’, taktub ‘she writes’, etc. (Arabic)

COMPOUNDING IN ENGLISH



MORPHOLOGICAL PROCESSES

- Inflection: modification of a word to express different grammatical categories (number, gender, tense, …)
  - dog → dogs
  - write → writes
- Derivation: process of forming a new word using an existing one. Changes meaning or part of speech.
  - happy → happiness
  - essen → essbar

SOME KINDS OF INFLECTION

- Case
- Number
- Clusivity
- Grammatical gender
- TAM
  - Tense
  - Aspect
  - Modality
  - Evidentiality
- Formality

QUESTIONS?

INTERLINEAR GLOSSES

How do linguists talk about languages that they do not speak?

- Interlinear morpheme-by-morpheme glosses are meant to give information about the meanings and grammatical properties of individual words and parts of words (interlinear: between the lines)
- Different levels of detail are chosen depending on the author's purposes and the readers' assumed background knowledge
- Example: Turkish sentence from earlier
  - Turkish: *Ben* *yüz-üyor-um*
    I.Nom *swim*-Prog-1P.Sg

THE LEIPZIG GLOSSING RULES

- Glosses need rules to be readable
- The famous standard rules for this are the **Leipzig Glossing rules**, developed in 2008 in Leipzig by
  - Bernard Comrie and Martin Haspelmath, at the Department of Linguistics of the Max Planck Institute for Evolutionary Anthropology
  - Balthasar Bickel, at the Department of Linguistics of Leipzig University
- 10 rules for how to gloss, and an appendix with a lexicon of proposed labels
- Examples of rules coming up (not for memorisation, just to get used to glossing and see more examples of morphological analysis)

RULE 1: WORD-BY-WORD ALIGNMENT

Interlinear glosses are left-aligned vertically, word by word, with the example

Indonesian (Sneddon 1996:237)

Mereka	*di*	*Jakarta*	*sekarang.*
they	in	Jakarta	now

'They are in Jakarta now.'

RULE 2: MORPHEME-BY-MORPHEME CORRESPONDENCE

Segmentable morphemes are separated by hyphens, both in the example and in the gloss. There must be exactly the same number of hyphens in the example and in the gloss.

Lezgian (Haspelmath 1993:207)

Gila	*abur-u-n*	*ferma*	*hamišaluğ*	*güğüna*	*amuq'-da-č.*
now	they-OBL-GEN	farm	forever	behind	stay-FUT-NEG

'Now their farm will not stay behind forever.'

RULE 3: GRAMMATICAL CATEGORY LABELS

Grammatical morphemes are generally rendered by abbreviated grammatical category labels, printed in upper case letters (usually small capitals). A list of standard abbreviations is in the appendix to the rules.

Russian

My	*s*	*Marko*	*poexa-l-i*	*avtobus-om*	*v*	*Peredelkino.*
1PL	COM	Marko	go-PST-PL	bus-INS	ALL	Peredelkino
we	with	Marko	go-PST-PL	bus-by	to	Peredelkino

'Marko and I went to Perdelkino by bus.'

RULE 4: ONE-TO-MANY CORRESPONDENCES

When a single object-language element is rendered by several metalanguage elements (words or abbreviations), these are separated by periods.

Turkish	Latin	French		German	
çık-mak	*insul-arum*	*aux*	*chevaux*	*unser-n*	*Väter-n*
come.out-INF	island-GEN.PL	to.ART.PL	horse.PL	our-DAT.PL	father.PL-DAT.PL
'to come out'	'of the islands'	'to the horses'		'to our fathers'	

Hittite (Lehmann 1982:211)

n=an	*apedani*	*mehuni*	*essandu.*
CONN=him	that.DAT.SG	time.DAT.SG	eat.they.shall

'They shall celebrate him on that date.' (CONN = connective)

Jaminjung (Schultze-Berndt 2000:92)

nanggayan	*guny-bi-yarluga?*
who	2DU.A.3SG.P-FUT-poke

'Who do you two want to spear?'

RULE 6: NON-OVERT ELEMENTS

If the morpheme-by-morpheme gloss contains an element that does not correspond to an overt element in the example, it can be enclosed in square brackets. An obvious alternative is to include an overt "Ø" in the object-language text, which is separated by a hyphen like an overt element.

Latin

puer	*puer-Ø*
boy[NOM.SG]	boy-NOM.SG
'boy'	'boy'

RULE 7: INHERENT CATEGORIES

Inherent, non-overt categories such as gender may be indicated in the gloss, but a special boundary symbol, the round parenthesis, is used.

Hunzib (van den Berg 1995:46)

oz#-di-g	*xõxe*	*m-uq'e-r*
boy-OBL-AD	tree(G4)	G4-bend-PRET

'Because of the boy the tree bent.'

(G4 = 4th gender, AD = adessive, PRET = preterite)

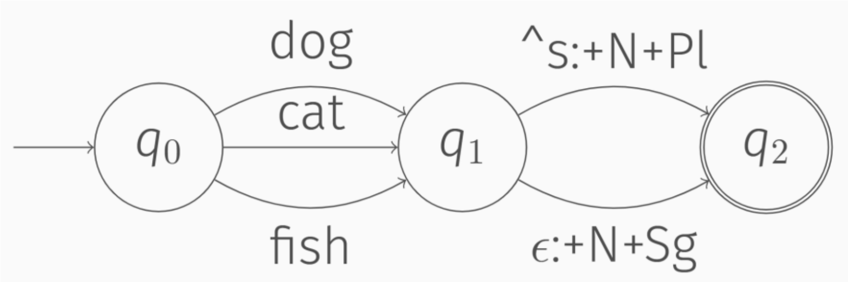
QUESTIONS?

COMPUTATIONAL MORPHOLOGY

FINITE STATE TRANSDUCERS FOR MORPHOLOGICAL ANALYSIS

Example: English noun plurals

LEXICAL	dog+N+Pl	dog+V+3+Sg+NPast	fish+N+Pl	fish+V+3+Sg+NPast
	↕	↕	↕	↕
MORPHEMIC	#dog^s#	#dog^s#	#fish^s#	#fish^s#
	↕	↕	↕	↕
SURFACE	dogs	dogs	fishes	fishes



DATASETS IN COMPUTATIONAL MORPHOLOGY

- Unimorph
- Morphological annotation in Universal Dependencies (more details in the next lecture on Syntax)

UNIMORPH: UNIVERSAL MORPHOLOGICAL ANNOTATION

UniMorph

Schema and datasets for universal morphological annotation

- Collaborative effort driven by SIGMORPHON, the ACL special interest group in Morphology
- 169 languages annotated in the UniMorph Schema
- 23 dimensions of meaning and over 212 features, but varying numbers of features, e.g.
  - 2 for finiteness: finite or infinite
  - 39 for case
- Source: mostly wiktionary

EXAMPLE OF UNIMORPH DATA

hablaste V;FIN;IND;PFV;PST;2;SG;INFM

Lemma hablar plus the feature bundle [FIN;IND;PFV;PST;2;SG;INFM] where

- FIN indicates a finite verb.
- IND indicates the indicative mood.
- PFV indicates the perfective aspect.
- PST indicates the past tense.
- 2 indicates the second person.
- SG indicates the singular number.
- INFM indicates the informal register.

SYNCRETISM

сказала V;FIN;IND;PFV;PST;1;SG;FEM

сказала V;FIN;IND;PFV;PST;2;SG;INFM;FEM

сказала V;FIN;IND;PFV;PST;3;SG;FEM

- Three entries for the same surface form in Russian, differing in the person feature
- Context is required to disambiguate, which is not provided by unimorph

MORPHOLOGICAL FEATURES IN UNIVERSAL DEPENDENCIES

- Morphological annotation in (primarily) syntactic treebanks → more on those next week
- Bottom-up approach: treebank annotators need a new feature, and the UD core group then consider if it occurs in at least one other language
- Lexical features are inherent to words, while inflectional features may depend on context

Lexical features*	Inflectional features*	
PronType	Gender	VerbForm
NumType	Animacy	Mood
Poss	NounClass	Tense
Reflex	Number	Aspect
Other	Case	Voice
Abbr	Definite	Evident
Typo	Deixis	Polarity
Foreign	DeixisRef	Person
ExtPos	Degree	Polite
		Clusivity

EXAMPLE UD FEATURE: MOOD

Mood: mood

Values:	Adm	Cnd	Des	Imp	Ind	Int	Irr	Jus	Nec	Opt	Pot	Prp	Qot	Sub

Mood is a feature that expresses modality and subclassifies finite verb forms.

- Universal feature list and guidelines on the website, but not all values are instantiated in all languages
- English-specific mood guidelines:

Ind: indicative

The indicative can be considered the d
attitude of the speaker.

Examples

- *He **makes** a sandwich.*

Imp: imperative

The speaker uses imperative to order

Examples

- ***Make** a sandwich!*

Sub: subjunctive

The subjunctive mood is used under c
an opinion or describing one's state of

Examples

- *I suggest that he **see** a doctor.*
- *If I **were** rich…*

QUESTIONS?

TASKS IN COMPUTATIONAL MORPHOLOGY

- Many were part of the SIGMORPHON series of shared tasks from 2016 to 2022
- Standard tasks
  - Morphological segmentation: segment a word into morphemes
  - Morphological analysis:
- Lemmatisation and stemming can also be considered computational morphology

SIGMORPHON 2016: MORPHOLOGICAL REINFLECTION

- Task 1: Inflection
  - Given a lemma with its POS, generated a target inflected form
  - Example:
    - Source lemma: `run`
    - Target tag: `Present participle`
    - Output: `running`
- Task 2: Reinflection
  - Given an inflected form and its current tag, generated a target inflected form. Task 2 is a harder case of Task 1, since the source tag is no longer guaranteed to be Lemma.
  - Example:
    - Source tag: Past
    - Source form: ran
    - Target tag: Present participle
    - Output: running
- Task 3: Unlabelled Reinflection
  - Given an inflected form without its current inflection, generate a target inflected form. Task 3 is a harder case of Task 2, since the source tag is no longer provided.
  - Example
    - Source tag: not given
    - Source form: ran
    - Target tag: Present participle
    - Output: running

SIGMORPHON 2017: PARADIGM CELL FILLING

Given a lemma, a part-of-speech tag, and an incomplete partial paradigm, complete the remaining paradigm cells. Note that for languages with smaller paradigms, it is often the case that no additional forms (other than the lemma) are observed.

```
Source lemma and part of speech: release V
```

Incomplete (covered) paradigm:

release	release	V;NFIN
release	--	V;3;SG;PRS
release	--	V;V.PTCP;PRS
release	released	V;PST
release	--	V;V.PTCP;PST

Target (uncovered) paradigm:

release	release	V;NFIN
release	releases	V;3;SG;PRS
release	releasing	V;V.PTCP;PRS
release	released	V;PST
release	released	V;V.PTCP;PST

SIGMORPHON 2018: INFLECTION IN CONTEXT

You are given a sentence with a number of missing word forms and the lemma of each missing word form. You are required to fill in the missing forms.

Track 1: lemmas and morphosyntactic descriptions provided for all context words

The/the+DT ___(dog) are/be+AUX;IND;PRS;FIN barking/bark+V;V.PTCP

Track 2: only forms provided for context words

The ___(dog) are barking

Target form:

dogs

SIGMORPHON 2019: CROSSLINGUAL TRANSFER

- Annotated resources, such as UniMorph data, for the world’s languages are not distributed equally.
- Some languages have more native speakers and willing and able to annotate more data
  - This doesn’t necessarily correlate with the number of speakers, or the amount of unlabelled data. See e.g. the overrepresentation of Swedish or Hebrew
- Cross-lingual transfer: using training data in one language to get better in another language

SIGMORPHON 2019: CROSSLINGUAL TRANSFER

Inflection generating, but with training data in a low-resource target language plus a high-resource source language. Testing is purely in the low-resource language

Low-resource target training data (Asturian)

```
facer    fechu    V;V.PTCP;PST
aguar    aguà     V;PRS;2;PL;IND
...
```

High-resource source language training data (Spanish)

```
tocar    tocando   V;V.PTCP;PRS
bailar   bailaba   V;PST;IPFV;3;SG;IND
mentir   mintió    V;PST;PFV;3;SG;IND
...
```

Test input (Asturian)

```
baxar    V;V.PTCP;PRS
```

Test output (Asturian)

```
baxando
```

SIGMORPHON 2019: MORPHOLOGICAL ANALYSIS AND LEMMATISATION IN CONTEXT

Given this sentence

```
# sent-id = 1
# text = They buy and sell books.
1       They    _       _       _       _       _       _       _       _
2       buy     _       _       _       _       _       _       _       _
3       and     _       _       _       _       _       _       _       _
4       sell    _       _       _       _       _       _       _       _
5       books   _       _       _       _       _       _       _       _
6       .       _       _       _       _       _       _       _       _
```

The system must produce

```
# sent-id = 1
# text = They buy and sell books.
1       They    they    _       _       N;NOM;PL        _       _       _       _
2       buy     buy     _       _       V;SG;1;PRS              _       _       _
3       and     and     _       _       CONJ            _       _       _       _
4       sell    sell    _       _       V;PL;3;PRS      _       _       _       _
5       books   book    _       _       N;PL            _       _       _       _
6       .       .       _       _       PUNCT           _       _       _       _
```

CONLL 2020: UNSUPERVISED MORPHOLOGICAL PARADIGM COMPLETION

- The set of all inflected forms of a lemma is called its paradigm
- While English does not have rich inflectional morphology, there are languages where a paradigm as over 1.5 million slots
- Children master the task of morphological inflection without access to explicit morphological information – can this be done by an automated system?

CONLL 2020: UNSUPERVISED MORPHOLOGICAL PARADIGM COMPLETION

The task: given the tokenised bible in the language, and a list of lemmas, complete the paradigm.

```
run
cooperate
sing
```

→

```
run         ran         1
run         runs        2
...
cooperate           cooperates        2
cooperate           cooperated        1
...
```

Crucially, one model is trained on a set of development languages, and then evaluated on the test languages that are revealed later → systems have to rely on universal properties of morphology

UPDATE ON SIGMORPHON SHARED TASKS

- 2021 and 2022 were left out of this lecture for time
- No more shared tasks after 2022: researchers generally moved on to LLM work
- This raises the question: what happens to computational morphology in the age of LLMs?

QUESTIONS?

MORPHOLOGY AND LLMS

IN THE AGE OF LLMS, WHY SHOULD WE STILL CARE ABOUT MORPHOLOGY?

- Clearly, LLMs are not explicitly modelling morphology in the way that FSTs were
- However, they're succeeding at morphology in the sense that they are clearly producing English that is morphologically well-formed
  - No nonsensical derivatives
  - Correct agreement between morphological features

RESEARCH QUESTION: HUMANLIKENESS

- Humans form abstractions about (morphological) processes in their language
- Language models produce perfect-looking text, but have they also formed abstractions, or are they simply memorising?

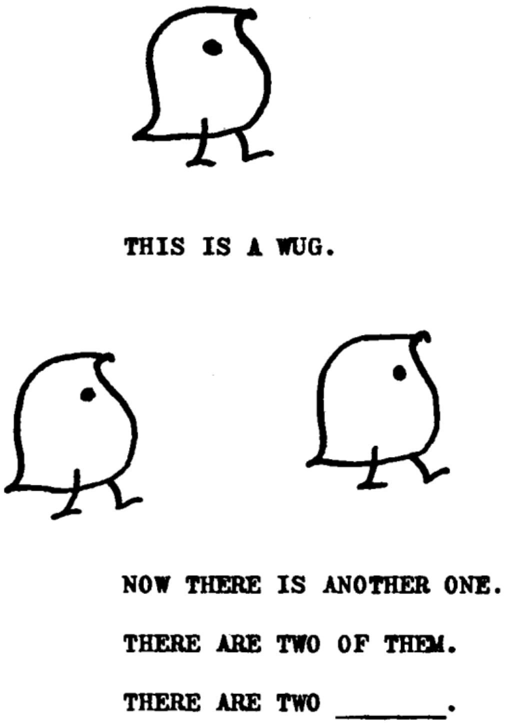
LLMS AND MORPHOLOGY

- Humans learn complex behaviour about language, that is more complex than a simple rule (e.g. with a regex) applied to all items
- Morphology is an ideal test bed for this: large groups of words (e.g. verbs) all undergoing the same change (e.g. past tense) with different outcomes (e.g. suffixes)

We know that **talk → talked**, but **teach → taught** and **swim → swam**

THE WUG TEST

1958: Jean Berko Gleeson tests if children have learned the general rule for how to pronounce the plural ‚s' depending on context, by making up fake words with specific properties and testing how the children respond

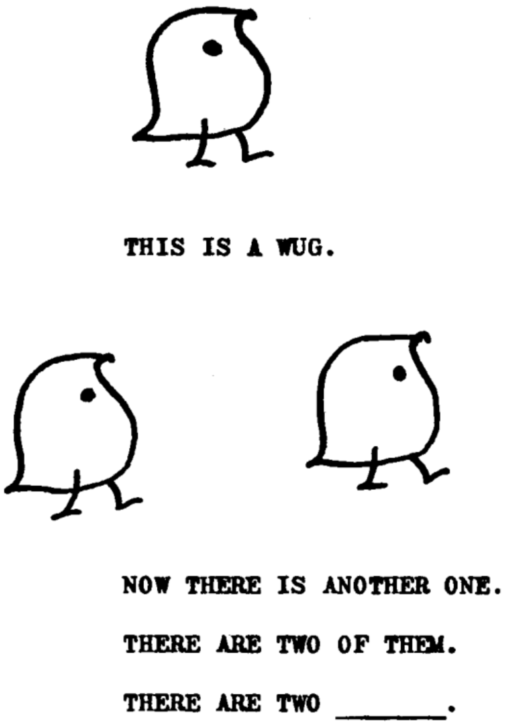


WUG-TESTING THE ENGLISH PAST TENSE

We can run a similar test for the English past tense

This is a man who knows how to SPOW. He is SPOWING.
He did the same thing yesterday. What did he do yesterday?
Yesterday he ________.

spowed ? spewed ?



DO LLMS FORM THE SAME GENERALISATIONS?

Sabine Doebel
@sabine_doebel

ChatGPT passes the wug test! (And with the enthusiasm of of a child who knows the answer!) But only inconsistently generalizes to other novel words @JeanBerkoG

Here is a wug. Now there are two. There are two ______.

Wugs!

Here is a dax. Now there are two. There are two ______.

Here is a dax. Now there are two. There are two dax.

Remi van Trijp @RemivanTrijp · Mar 2, 2023
Wugs have almost certainly been in its training data, so it is already known by it. You have to invent a new set of promps to test it properly

1 1 160

Sabine Doebel @sabine_doebel · Mar 2, 2023
That's what I figured. I used some made up ones and it was inconsistent.

1 100

MMitchell
@mmitchell_ai

Um, has anyone succeeded in replicating something like the "wug" study with ChatGPT? I've been trying for the past 20 minutes and I already have a headache.
I think it might fail from the very start.

MA I'm going to try to teach you some new words, and then I'd like to ask you some questions.

Sure, I'm always happy to learn new words and answer your questions. Please go ahead!

MA Okay, the first new word is "glack".

I'm sorry, but "glack" doesn't seem to be a real word in the English language. Could you provide some context or a definition for it?

PROMPTS IN ALL LANGUAGES

English:

- Long: “Fill in the blank with the correct past tense of the verb X. Answer with one word. They X all the time. In fact, they _ just yesterday! _ :”
- Short: “Form the correct past tense of the verb X. Answer with one word. X :”

German:

- Long: “Fülle die Lücke mit dem korrekten Plural des Nomens X aus. Antworte mit einem Wort. Hier ist ein X. Jetzt sind es zwei _! _:”
- Short: “Bilde den korrekten Plural des Nomens X. Antworte mit einem Wort. X :”

Tamil:

- Long: “நேற்று அவரிடம், "நீ X" என்றேன். அதைக் கேட்டு அவன் போய் _. _:”
- Short: “X :”

Turkish:

- Long: “Boşlukları X ile verilen eylemin birinci tekil şahıs geçmiş zaman formları ile doldurun. Ben her zaman X. Ama dün _. _:”
- Short: “Tek bir sözcük ile farazi X eyleminin birinci tekil şahıs geçmiş zaman hali nasıl olur? X :”

EXPERIMENTAL SETUP

Long prompt

zero-shot

one-shot

few-shot

Fill in the blank with the correct past tense of the verb "bebit". Answer with one word.
They bebit all the time. In fact, they ___ just yesterday!
test : tested
teach : taught
build : built
sing : sang
hit : hit
___ :

The verb "bebit" is a made-up word, so its past tense doesn't follow standard English conjugation rules. Based on the pattern in your examples, it appears that you're asking for an irregular past tense form.

For consistency with the examples like "hit: hit," where the base and past tense forms are the same, the past tense of "bebit" could simply be "bebit."

So the answer would be:

bebit: bebit

Short prompt

zero-shot

one-shot

few-shot

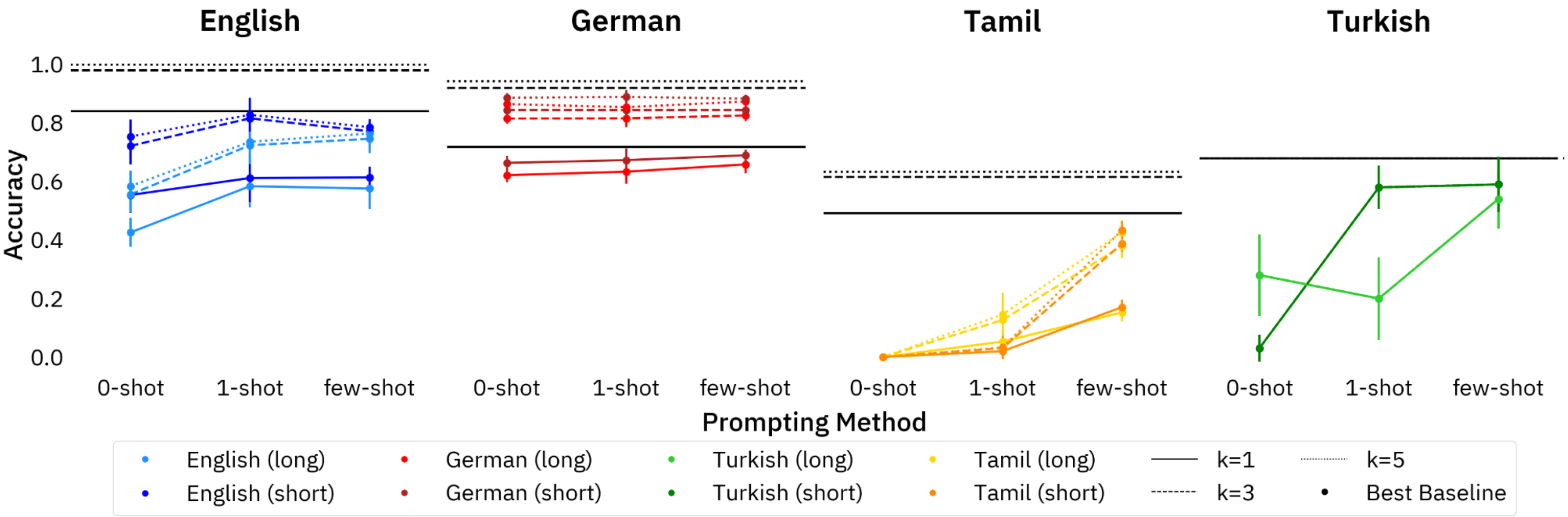
Form the correct past tense of the verb "bebit". Answer with one word.
test: tested
teach: taught
build: built
sing: sang
hit: hit

The correct past tense of "bebit" is "bebet."

RESULTS

bebitted	x 13
bebited	x 10
bebit	x 2
bebited	x 1
bebat	x 1

Crucially, humans don't all agree, and the ideal language model should either agree with the majority, or even represent the full distribution



OVERGENERALISATION

- German few-shot results by noun ending
- Main source of error is overgeneralisation morphemes, +*en* and +*s*

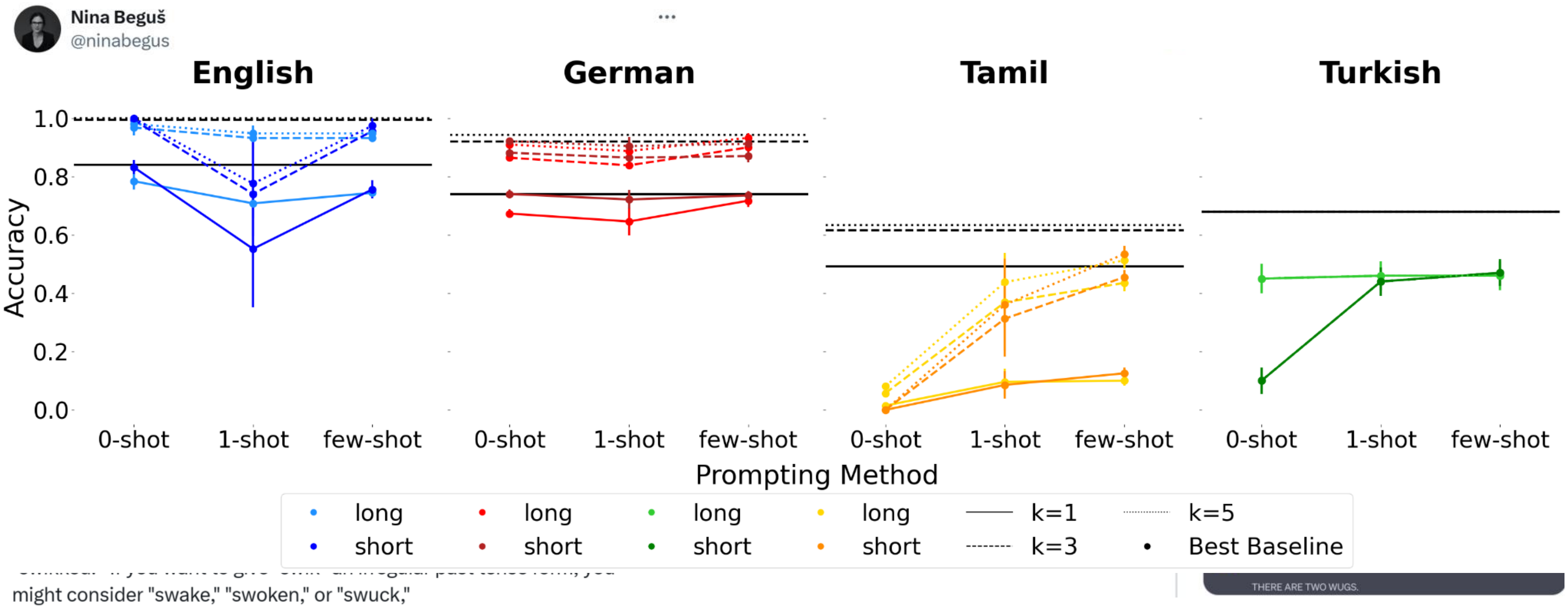
→ Amplification of training data biases

Gold \ Predicted	No change	+ e	+ en	+ er	+ s	Vowel change	Vowel change + e	Vowel change + er	Real Word	Unknown
No change	317	6	0	0	10	0	0	0	9	8
+ e	47	262	77	5	79	0	13	5	22	19
+ en	3	20	64	0	18	4	0	0	0	11
+ s	25	15	5	0	109	0	1	0	1	4
Vowel change + e	0	13	0	0	1	0	22	3	0	1
Unknown	7	3	0	0	13	0	0	0	0	25

CAVEAT: INSTRUCTION FOLLOWING

- Prompts without original wug-context performed better
- Classic wug-question produces no parseable results
- For Tamil any reply required prompt-tuning and arguing
- Even for English: *bebit* → *drained, drank, drank, bebitted, bebit, bebit, bet, bebitted, bebit, bdrank*

UPON POPULAR REQUEST: REPEAT WITH GPT-4



CONCLUSIONS DRAWN

- GPT4, and probably all larger models, form similar abstractions about inflectional morphology as humans, at least for English
- This means that it is possible for a neural network to learn the underlying patterns purely from data → what does that mean for linguistics?
- But for most languages, LLMs aren't even close to modelling their morphology
  - Too little data?
  - Or because most languages are morphologically more complex than English?

QUESTIONS?

RESEARCH QUESTION: TRANSFORMER ARCHITECTURE AND MORPHOLOGY

- LLMs process subword tokens, not morphemes
- Subword tokens may sometimes correspond to morphemes
- Questions
  - To what degree and for which languages do they correspond to morphemes?
  - And does that even matter for LLM performance?

RECALL HOW BPE WORKS

$R:$

k	(t_L, t_R)
1:	$(\texttt{o},\ \texttt{r})$
2:	$(\texttt{␣h},\ \texttt{or})$
3:	$(\texttt{␣hor},\ \texttt{s})$
	$\vdots$

$$V = \{\ \texttt{,},\ \texttt{!},\ \texttt{␣a},\ \texttt{␣f},\ \texttt{␣h},\ \texttt{␣k},\ \texttt{␣m},\ \texttt{A},\ \texttt{d},\ \texttt{e},\ \texttt{f},\ \texttt{g},\ \texttt{h},\ \texttt{i},\ \texttt{k},\ \texttt{m},\ \texttt{n},\ \texttt{o},\ \texttt{r},\ \texttt{s},\ \texttt{y},\ \texttt{dom},\ \texttt{␣for},\ \texttt{␣hor},\ \texttt{␣hors},\ \texttt{␣horse},\ \texttt{␣king},\ \texttt{␣my},\ \texttt{or},\ \ldots\}$$

$$I = \{\ (\texttt{A}; 1),\ (\texttt{␣a}; 2),\ (\texttt{␣for}; 1),\ (\texttt{␣horse}; 3),\ (\texttt{␣king},\texttt{dom}; 1),\ (\texttt{␣my}; 1),\ (\texttt{,}; 2),\ (\texttt{!}; 1)\ \}$$

$$d': ((\texttt{A}), (\texttt{␣h}, \texttt{o}, \texttt{r}, \texttt{s}, \texttt{e}), (\texttt{,}), (\texttt{␣a}), (\texttt{␣h}, \texttt{o}, \texttt{r}, \texttt{s}, \texttt{e}), (\texttt{,}), (\texttt{␣m}, \texttt{y}), (\texttt{␣k}, \texttt{i}, \texttt{n}, \texttt{g}, \texttt{d}, \texttt{o}, \texttt{m}), (\texttt{␣f}, \texttt{o}, \texttt{r}), (\texttt{␣a}), (\texttt{␣h}, \texttt{o}, \texttt{r}, \texttt{s}, \texttt{e}), (\texttt{!}))$$

$$1: ((\texttt{A}), (\texttt{␣h}, \texttt{or}, \texttt{s}, \texttt{e}), (\texttt{,}), (\texttt{␣a}), (\texttt{␣h}, \texttt{or}, \texttt{s}, \texttt{e}), (\texttt{,}), (\texttt{␣m}, \texttt{y}), (\texttt{␣k}, \texttt{i}, \texttt{n}, \texttt{g}, \texttt{d}, \texttt{o}, \texttt{m}), (\texttt{␣f}, \texttt{or}), (\texttt{␣a}), (\texttt{␣h}, \texttt{or}, \texttt{s}, \texttt{e}), (\texttt{!}))$$

$$2: ((\texttt{A}), (\texttt{␣hor}, \texttt{s}, \texttt{e}), (\texttt{,}), (\texttt{␣a}), (\texttt{␣hor}, \texttt{s}, \texttt{e}), (\texttt{,}), (\texttt{␣m}, \texttt{y}), (\texttt{␣k}, \texttt{i}, \texttt{n}, \texttt{g}, \texttt{d}, \texttt{o}, \texttt{m}), (\texttt{␣f}, \texttt{or}), (\texttt{␣a}), (\texttt{␣hor}, \texttt{s}, \texttt{e}), (\texttt{!}))$$

$$3: ((\texttt{A}), (\texttt{␣hors}, \texttt{e}), (\texttt{,}), (\texttt{␣a}), (\texttt{␣hors}, \texttt{e}), (\texttt{,}), (\texttt{␣m}, \texttt{y}), (\texttt{␣k}, \texttt{i}, \texttt{n}, \texttt{g}, \texttt{d}, \texttt{o}, \texttt{m}), (\texttt{␣f}, \texttt{or}), (\texttt{␣a}), (\texttt{␣hors}, \texttt{e}), (\texttt{!}))$$

$$\vdots$$

$$k: ((\texttt{A}), (\texttt{␣horse}), (\texttt{,}), (\texttt{␣a}), (\texttt{␣horse}), (\texttt{,}), (\texttt{␣my}), (\texttt{␣king}), (\texttt{dom}), (\texttt{␣for}), (\texttt{␣a}), (\texttt{␣horse}), (\texttt{!}))$$

BPE SOMETIMES YIELDS MORPHEME-LIKE SEGMENTS, BUT OFTEN DOES NOT

BPE operates blindly

	peed	*deed*
Linguist_1	pe@@ ed	deed
Linguist_2	pee@@ d	deed
BPE_1	pe@@ ed	de@@ ed
BPE_2	peed	deed

THE LEFT-TO-RIGHT NATURE OF BPE

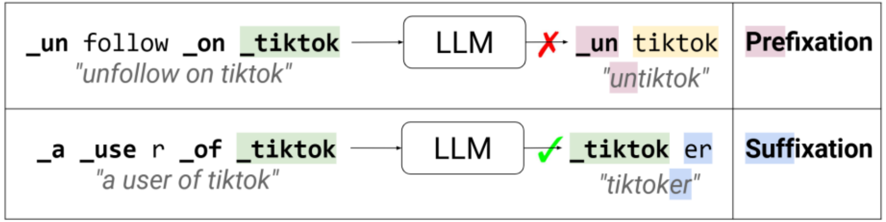


Figure 1: In BPE tokenization, marking word-initial tokens with "_" hinders the generation of prefixed forms (e.g., "`_un tiktok`"), as they do not share any token with their base (e.g., "`_tiktok`"). Identical tokens are highlighted in the same color.

- BPE distinguishes if tokens are word-initial or not
- This means that roots are different tokens when prefixes attach to them
- They become worse at prefixation, probably because of this

SOME LANGUAGES ARE PROBABLY BETTER SERVED BY BPE

MorphScore: how well does the morphological segmentation match the BPE segmentation?

Language	Word	Source	Segmentation	Score
Basque	aldiz	morphemic	aldi + z	
		Tokenizer 1	[‘al’, ‘diz’]	0
		Tokenizer 2	[‘aldi’, ‘z’]	1
Croatian	suučesnika	morphemic	suučesnik + a	
		Tokenizer 1	[‘su’, ‘uče’, ‘s’, ‘nika’]	0
		Tokenizer 2	[‘su’, ‘u’, ‘če’, ‘s’, ‘nika’]	0
Icelandic	samráðs	morphemic	samráð + s	
		Tokenizer 1	[‘samráð’, ‘s’]	1
		Tokenizer 2	[‘samráðs’]	exclude
Greek	Αδριανής	morphemic	Αδριανή + ς	
		Tokenizer 1	[‘Α’, ‘δ’, ‘ριανής’]	0
		Tokenizer 2	[‘Α’, ‘δρ’, ‘ιανή’, ‘ς’]	1

MORE MORPHOLOGICALLY COMPLEX LANGUAGES HAVE WORSE PERFORMANCES IN LLMS…BUT WHY?

- Does a worse match between morphemes and BPE really lead to worse language modelling?
- Or do they have other properties that will always make them harder to model?
- Or do they are lower-resource in general?

→ This remains an open question

QUESTIONS?